Harshawardhan Ramesh Kapadnis

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AI/ML & Full Stack Developer with a demonstrated aptitude for translating applied research into production-grade systems. Adept at operating across the complete development lifecycle architecting intelligent models, engineering the surrounding infrastructure, and delivering functional, deployable products. Distinguished by a systems-oriented approach that prioritizes robustness, scalability, and real-world applicability over isolated model performance.

SKILLS

Programming & Web Development:	Python, JavaScript, React.js, Node.js, , RESTAPIs, Backend
AI/ML & Generative AI:	TensorFlow, OpenCV, Computer Vision, CNN, NLP, RAG, n8n model , Gen AI
Databases & Tools:	SQL, MongoDB, PostgreSQL, Git, GitHub, Postman, VS Code

EXPERIENCE

Gen AI Intern at Combat Solutions Dec 2025 - Jun 2026

P2X Maker: P&ID-to-DCS Converter

Tech:Python, YOLOv8, FastAPI, React, Computer Vision, PaddleOCR, Neo4j, SQLite, Draw.io, XAML.

- Engineered an automated P&ID to DCS conversion system using YOLOv8 segmentation with intelligent patch-based tiling to detect 100+ industrial symbols and pipe connections, eliminating weeks of manual digitization work.
- Developed three-method line detection combining CLAHE, Canny edge detection, and adaptive thresholding for accurate pipe and connection extraction across varying drawing styles and quality levels.
- Built a full-stack React and FastAPI platform with integrated Draw.io editor, PaddleOCR text extraction, and Neo4j connectivity mapping for seamless symbol review, correction, and multi-format export (XAML, SVG, JSON).

Safe Driver Recognition System

Tech: Python, OpenCV, Deep Learning, Twilio API.

- Implementedblink detection andface trackingfor drowsiness detection.
- Integrated Twilio API for real-time alert notifications.
- Improved detection accuracy through modeltuning and performance optimization.
- Integrated emotion detection and heart rate monitoring to identify driver emotions and potential cardiac events, with real-time Twilio notifications for emergency alerts.

PROJECTS

Blockchain Based Employee Reward Management System

TechStack:MERNStack, Solidity,EthereumBlockchain, Web3.js.

- Built a blockchain-based employee rewarding platform for automated performance evaluation and smart contract-based reward allocation.
- Enabled secure and tamper-proof on-chain reward tracking, reducing manual HR effort and eliminating bias.
- Developed admin and employee dashboards for task management, reward redemption, and immutable record verification.

Deepfake Detection System - Media Authenticator

TechStack: TensorFlow, OpenCV, Python, CNN,Deep Learning, Computer Vision.

- Built a CNN-based deepfake detection system achieving 92% accuracy on 10,000+ images for reliable real-time media verification.
- Developed high-speed preprocessing pipelines processing 500+ images/min, enabling scalable and faster deepfake detection.
- Optimized model performance by reducing false detections by 15% across multiple datasets for improved reliability.

ACHIEVEMENTS

National Finalist - Smart India Hackathon 2024

- Developed a vacuum - insulated , pressure - regulated container reducing nitrogen loss by 40% for scalable livestock semen preservation.

3rd Place - IoT Project Competition 2023

- Built an automated solar tracking system using LDR sensors to improve solar energy efficiency.

EDUCATION :

K.K. Wagh Institute of Education and Research, Nashik Nov2022 – Jun 2026

B.Tech in Artificial Intelligence & DataScience **CGPA : 7.80**